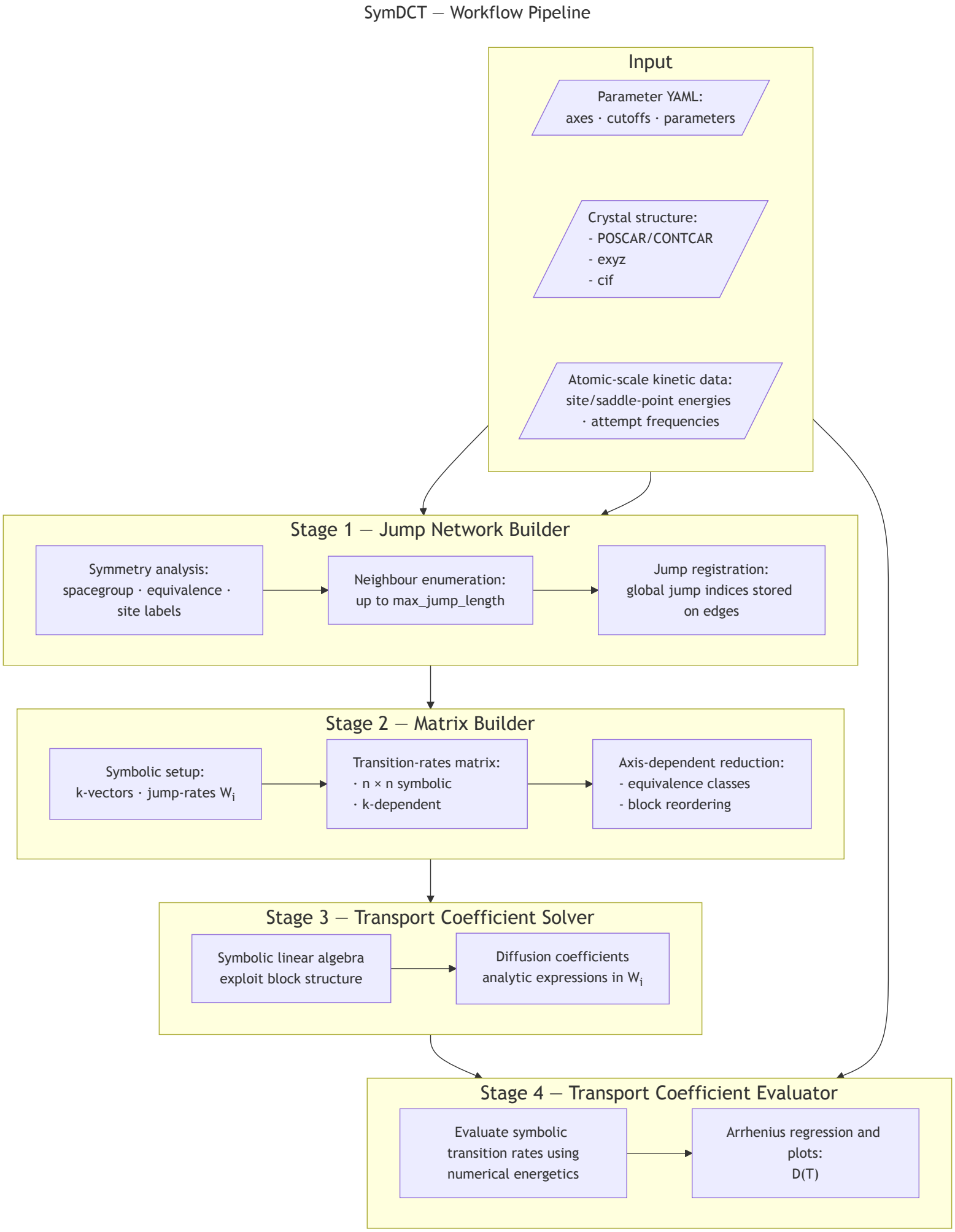

SymDCT — Workflow Pipeline
Input
Parameter YAML:
axes · cutoffs · parameters
Crystal structure:
- POSCAR/CONTCAR
- exyz
- cif
Atomic-scale kinetic data:
site/saddle-point energies
· attempt frequencies
Stage 1 — Jump Network Builder
Symmetry analysis:
spacegroup · equivalence ·
site labels
Neighbour enumeration:
up to max_jump_length
Jump registration:
global jump indices stored
on edges
Stage 2 — Matrix Builder
Symbolic setup:
k-vectors · jump-rates W_i
Transition-rates matrix:
· n × n symbolic
· k-dependent
Axis-dependent reduction:
- equivalence classes
- block reordering
Stage 3 — Transport Coefficient Solver
Symbolic linear algebra
exploit block structure
Diffusion coefficients
analytic expressions in W_i
Stage 4 — Transport Coefficient Evaluator
Evaluate symbolic
transition rates using
numerical energetics
Arrhenius regression and
plots:
D(T)